

Software



CodeHS



What is
software?

Software

- Also called “programs” or “applications”
- Runs on the computer
- Contains instructions to tell the computer exactly what to do

User Application Software

Programs that run
on the computer
that help users
perform tasks.

User Applications

- Programs that are installed onto your computer by you
- Enables the user to accomplish tasks using the application
- Written by programmers, but packaged so that other people can install it on their own computers

Application Examples

- Games
- Office applications
 - Word processors
 - Spreadsheet applications
- Web browser
- File explorer

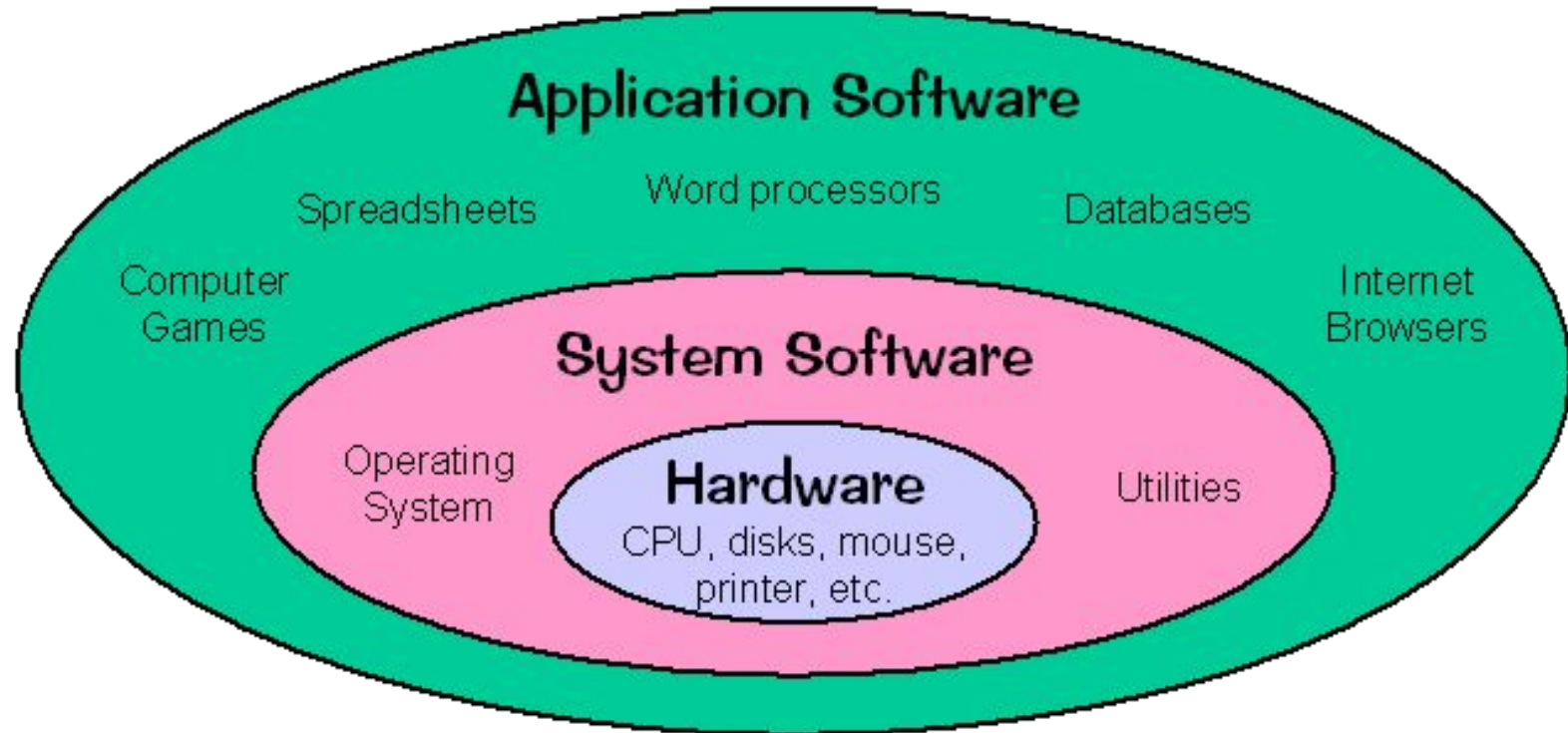


Common Web Browsers

System Software

Programs that run
on the computer
that help the
computer perform
tasks.

System Software?



System Software

- Layer between User Applications and Hardware
- User does not directly interact with it

Examples

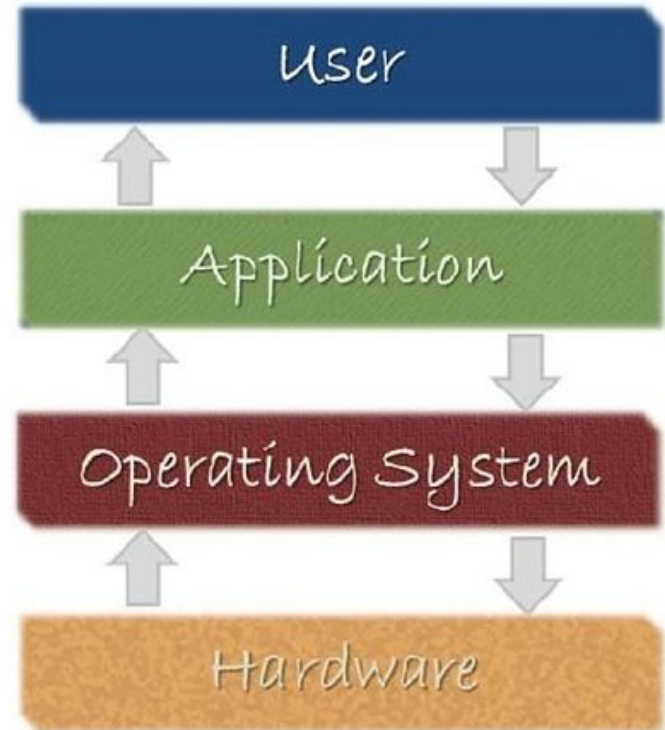
- **Boot program**
 - Loads the operating system into main memory when the computer turns on
- **BIOS**
 - Basic input/output system
 - Manages communication between input/output devices and the operating system
- **Device Driver**
 - Software for a specific input/output device

Operating Systems

Makes your
computer usable.

Where Does the Operating System Live?

- Between User Applications and Hardware
- Is one type of system software



What Does the OS Do?

- Operating System also known as OS
- Manages user applications
- Manages memory allocation among applications
- Manages multi-tasking
- Displays a Graphical User Interface (GUI)
 - Enables you to interact with the computer using application displays and icons rather than a text-based console

Common Operating Systems

- Microsoft Windows
- Apple Mac OS (OS X)
- Linux
- Unix



Other Computer Programs

A program can also refer to a small piece of code that the user writes.

Program

- A set of specific instructions for the computer written by a programmer using a programming language
- Written to perform a task, but is not meant to be installed as software on other computers

Programming Language

- Human readable way to write computer instructions



Some programming languages ^[1]

Sources

Pictures

- <https://pixabay.com/>
- <http://courses.cs.vt.edu/csonline/OS/Lessons/Introduction/index.html>
- [1]By Wiziq Inc. [Public domain], via Wikimedia Commons

Content

- <http://whatis.techtarget.com/definition/system-software>
- <http://whatis.techtarget.com/definition/operating-system-OS>